TÓM TẮT TIẾNG ANH CHUYÊN NGÀNH (TACN)

BOOK 1

Chương 2 - Introduction to Computing Systems

Phân loại máy tính:

Type / Khái niệm	Description (Mô tả)	Used For / Ví dụ
Desktop	A personal computer designed to be used at a single location.	Hybrid console, PC, Handheld, PDAs
Microcomputer	Term for a PC; includes desktops, laptops, etc.	Nó cũng là máy tính cá nhân vs laptop
Minicomputer	Medium-sized computer; stronger than micro, weaker than mainframe.	- Industrial control
Workstation	High-performance desktop for specialized tasks.	- Multi-user systems - Graphic design & video editing - 3D modeling and animation - Software development - Scientific research
Server	Provides data, services, or resources over a network.	- Managing business operations - Data collection and analysis - Hosting websites & databases - File storage/sharing - Email/cloud services - User access control
Mainframe	Very large, powerful computer for large organizations.	- Transaction processing (e.g., banking) - Big data storage - Enterprise apps - Airline/insurance systems

Chương 3 - Inside the Computer

Từ viết tắt về các linh kiện và nhận diện:

Viết tắt	Tên đầy đủ (Full term)	Từ nhận diện / Chức năng
CPU	Central Processing Unit	Processor / Brain of the computer

RAM	Random Access Memory	Temporary storage
ROM	Read-Only Memory	Permanent instructions
ALU	Arithmetic Logic Unit	Calculations and logic
CU	Control Unit	Coordination and control
USB	Universal Serial Bus	Peripheral connection
GHz	Gigahertz	CPU speed
BIOS	Basic Input/Output System	Hardware check
HDD	Hard Disk Drive	Mechanical storage
SSD	Solid State Drive	Fast storage
PSU	Power Supply Unit	Power provider
GPU	Graphics Processing Unit	Graphic processing
MB	Motherboard	Component hub

Vocabulary:

- **Heat sink:** a part used to cool the processor.
- **Expansion card:** a part used to increase a computer's functions.
- **Processor:** a part that completes tasks for the computer.
- **Hard drive:** a part used to store large amounts of data.
- **Motherboard:** a slotted board with circuits and ports.

Chương 4 - Computing Devices

(Chương này thêm các thiết bị input/output cơ bản nên đã được chọn lọc)

Từ viết tắt và nhận diện:

Viết tắt	Tên đầy đủ	Từ nhận diện / Chức năng
Wi-Fi	Wireless Fidelity	Wireless network connection
LCD	Liquid Crystal Display	Thin display screen
LED	Light Emitting Diode	Energy-saving screen
IoT	Internet of Things	Smart connected devices
GPS	Global Positioning System	Location tracking
OS	Operating System	Hardware-software interface
AI	Artificial Intelligence	Smart automation / virtual assistant

Các loại bộ nhớ:

Loại bộ nhớ	Vai trò (Role)	Các thiết bị tiêu biểu
Primary Storage	Temporarily stores data being processed. - Very fast access speed.	- RAM (Random Access Memory) - ROM (Read-Only Memory) - Cache Memory

		- CPU Registers
Secondary Storage	Stores data permanently.	- HDD (Hard Disk Drive)
	- Retains data after power off.	- SSD (Solid State Drive)
		- SSHD (Hybrid Drive)
Auxiliary/External Storage	External storage devices.	- USB flash drive
	- Used for backup or data transfer.	- Memory card (SD card, microSD)
		- External HDD/SSD
		- CD/DVD/Blu-ray
Tertiary/Archive Storage	Used for large-scale, long-term storage.	- Magnetic tape (Tape Drive)
	- Infrequently accessed.	- Optical jukebox
		- Archival cloud storage (e.g., Amazon Glacier)

Vocabulary:

- **Keyboard:** a device that enters information into a computer.
- **Mouse:** a device is used to control a pointer or cursor.
- **Scanner:** a device that reads images and sends them to a computer.
- **Microphone:** a device that gathers sounds.
- **Monitor:** a device that displays images.

Chương 5 - Networking

- Key facts:
  - A computer on the Internet is uniquely identified by an **IP address**.
  - **ISP** is a company that provides users with access to the internet.

Viết tắt và nhận diện các từ của chương:

Viết tắt	Tên đầy đủ	Từ nhận diện
LAN	Local Area Network	Local network (home, office)
WAN	Wide Area Network	Large-scale network
MAN	Metropolitan Area Network	City-scale network
IP	Internet Protocol	Device Identification, data routing
URL	Uniform Resource Locator	Website address
HTTP	HyperText Transfer Protocol	Web communication
HTTPS	HTTP Secure	Secure web communication
ISP	Internet Service Provider	Internet provider
Wi-Fi	Wireless Fidelity	Wireless internet

DSL	Digital Subscriber Line	Internet over phone line
IP Address	Internet Protocol Address	Network device identifier
DNS	Domain Name System	Domain to IP translation
MAC	Media Access Control	Hardware address
VPN	Virtual Private Network	Secure online connection
TCP/IP	Transmission Control Protocol / Internet Protocol	Internet data transmission protocol

Phân biệt các độ lớn của các loại mạng:

Abbreviation	Name	Phạm vi	Typical Use Cases
LAN	Local Area Network	Nhỏ - trong phòng / tòa nhà	Internal office network, classrooms, home networks
WLAN	Wireless Local Area Network	Giống LAN nhưng không dây (Wi-Fi)	Home Wi-Fi, cafés, wireless office setups
MAN	Metropolitan Area Network	Trung bình - phạm vi thành phố	Connecting offices across a city
WAN	Wide Area Network	Rất rộng – liên tỉnh, liên quốc gia	Global enterprise networks, bank systems
Internet	Global Network	Lớn nhất, toàn cầu	Accessing websites, email, social media, video streaming
Intranet	Private Internal Network	Nội bộ - bên trong một tổ chức	Internal company systems, document sharing
Extranet	Extended Internal Network	Mở rộng Intranet có kiểm soát	Secure partner/vendor access to company internal resources

(Lưu ý: 2 cái cuối là Intranet và Extranet phân biệt dựa trên quyền truy cập, không phải độ lớn địa lý).

Vocabulary:

- The **router** lets a network connect to the Internet.
- A **LAN** is a type of network.
- Businesses communicate on the **Internet**.
- We do not use cables for our network. We have a **WLAN**.
- **Routers** need a correct configuration to work properly.
- Businesses with a LAN use **CAT-5 cable** to connect computers.
- It is sometimes necessary to **cycle** a network to make it work.
- The **DHCP** provides different IP addresses for computers.

Chương 6 - The User Interface

Viết tắt	Tên đầy đủ	Từ nhận diện / Chức năng
GUI	Graphical User Interface	Graphic interface (icons, windows, menus)
CLI	Command Line Interface	Command input via keyboard
UX	User Experience	User experience
UI	User Interface	User interaction interface

OS	Operating System	System controller

Vocabulary:

- Use the **pointer** to choose files and commands.
- Small pictures or symbols in software are **icons**.
- The **operating system** is the program that makes a computer work.
- When something is on the **desktop** you see it onscreen.
- An OS that uses small pictures to represent files is a **GUI**.
- You can **minimize** files and programs in different windows.

Chương 7 - Word Processing

Viết tắt	Tên đầy đủ	Function / Role Keyword
WYSIWYG	What You See Is What You Get	Visual editing interface (Word, Google Docs)
RTF	Rich Text Format	Rich text formatting (bold, italic, colors)
DOCX	Microsoft Word Document	Microsoft Word file format
PDF	Portable Document Format	Fixed-layout document, view-only

Vocabulary:

- Use a **macro** to type out phrases that you use a lot.
- My file won't open in other programs due to a **compatibility** problem.
- Instead of writing by hand, use a **word processing program**.
- This file is in a **format** that doesn't open in TypeWrite.
- **Save** each document with a different name so you have a copy of each.
- The letter's **font** is wrong. The heading is in the wrong place.
- Increase the font **size** to make the file easier to read.
- Most word processors offer **spell check** to help you find mistakes.

Chương 8 - Email

- The **Subject** is a brief description of what the email is about.

Viết tắt	Tên đầy đủ	Function / Role Keyword
Email	Electronic Mail	Digital message system
SMTP	Simple Mail Transfer Protocol	Sending emails
POP3	Post Office Protocol version 3	Downloading emails
IMAP	Internet Message Access Protocol	Email syncing across devices
ISP	Internet Service Provider	Internet provider
CC	Carbon Copy	Send visible copy
BCC	Blind Carbon Copy	Send hidden copy
URL	Uniform Resource Locator	Embedded web link

Vocabulary:

- A computer that sends email: **email client**
- A program that saves, sends and receives email: **SMTP server**

- Words/letters used to identify companies and addresses: **domain name**
- Letters/numbers that let a person access something: **password**
- A file that is sent with an email: **attachment**

Chương 9 - Web Browsing

Vocabulary:

- **Hyperlink:** a word/image that you can click on to navigate.
- **Website:** a collection of related Internet pages.
- **HTML:** a computer language used to make web sites.
- **Web host:** a company that provides space for a website.
- **Web browser:** a program that lets you look at web pages.

BOOK 2

Unit 1 - Computer Memory

Vocabulary:

- Where data that doesn't fit in the RAM is stored: **virtual memory**
- The main circuit board in a computer: **motherboard**
- A unit of measurement that measures the speed of a CPU: **MHz**
- A data storage method that stores data on the hard drive: **paging file**
- The smallest unit of computer data: **bit**
- A memory module that sends up to 32 bits of data to a CPU: **SIMM** (Single Inline Memory Module)
- A memory module that sends up to 64 bits of data to a CPU: **DIMM** (Dual Inline Memory Module)

Unit 2 - Programming Languages

Vocabulary:

- **Compiler:** a language that computers can process / A computer part that changes source code into machine code.
- **Java:** An object-oriented programming language used often.
- **Assembler:** A programming language that translates between programs.
- **BASIC:** An early programming language that is not widely used anymore.
- **C#:** A competitor to Java.
- **C++:** A programming language that adds capabilities to C.
- **C:** Used to program Linux.
- **CPU:** Central Processing Unit.

Các thuật ngữ khác:

Viết tắt	Tên đầy đủ (Full từ)	Nhận diện
IDE	Integrated Development Environment	Programming tool, editor, compiler
OOP	Object-Oriented Programming	Class, object, method, inheritance

Unit 3 - ISPs and Internet Access

Vocabulary:

- **T-1:** lines that allow many users to connect to the Internet.
- **Wi-Fi:** an Internet connection using wireless signals.

- **ISP:** a company that provides access to the Internet.
- **Wireless router:** a device that routes data wirelessly.
- **Broadband:** a system allowing the transfer of large amounts of data.
- **DSL (Digital Subscriber Line):** high-speed internet via telephone line.

Các Unit còn lại của Book 2 (Tiêu đề)

- **Unit 4:** Storage Devices
- **Unit 5:** Peripherals
- **Unit 6:** MP3 Players
- **Unit 7:** Cell Phones
- **Unit 8:** Communications
- **Unit 9:** Entertainment and Social Networking
- **Unit 10:** GPS Systems
- **Unit 11:** Banking
- **Unit 12:** Education and Research
- **Unit 13:** Robotics
- **Unit 14:** Electronic Publishing
- **Unit 15:** Artificial Intelligence